



Cross-document Event Identity via Dense Annotation

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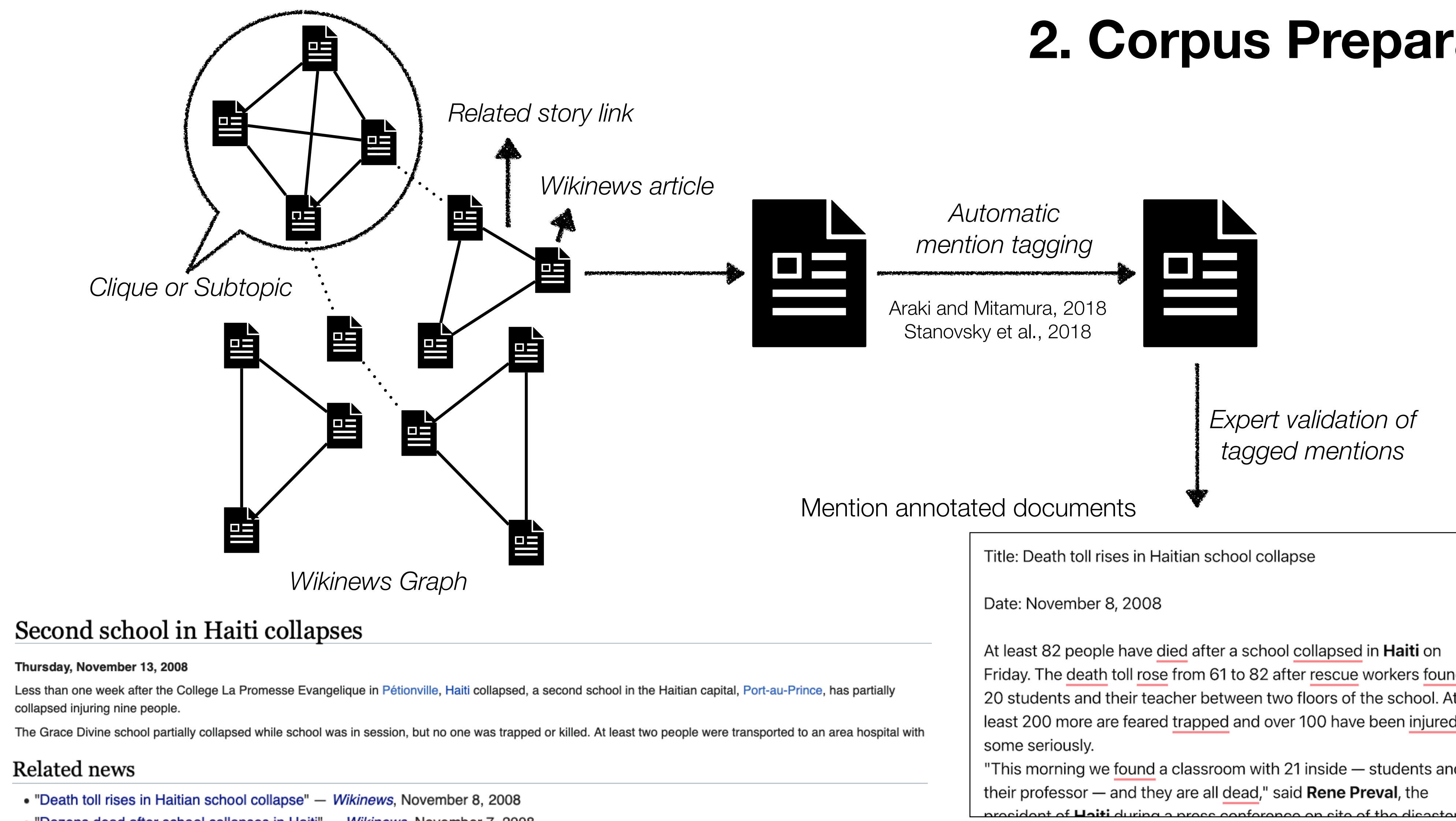
github.com/adithya7/cdec-wikinews

1. CDEC-WN

- A new dataset for cross-document event coreference on English Wikinews
- Proposed annotation framework tackles partial identity of mentions (Recasens et al., 2011, Hovy et al., 2013)
- Consists of a dense annotation of mention spans by considering open-domain events
- Presents evidence for a new type of partial identity, spatiotemporal continuity

# topics	1
# subtopics	55
# documents	176
# document pairs	198
# event mentions	7220
# CDEC links	4282
# CDEC links per doc.	21.6
# annotators	46

2. Corpus Preparation

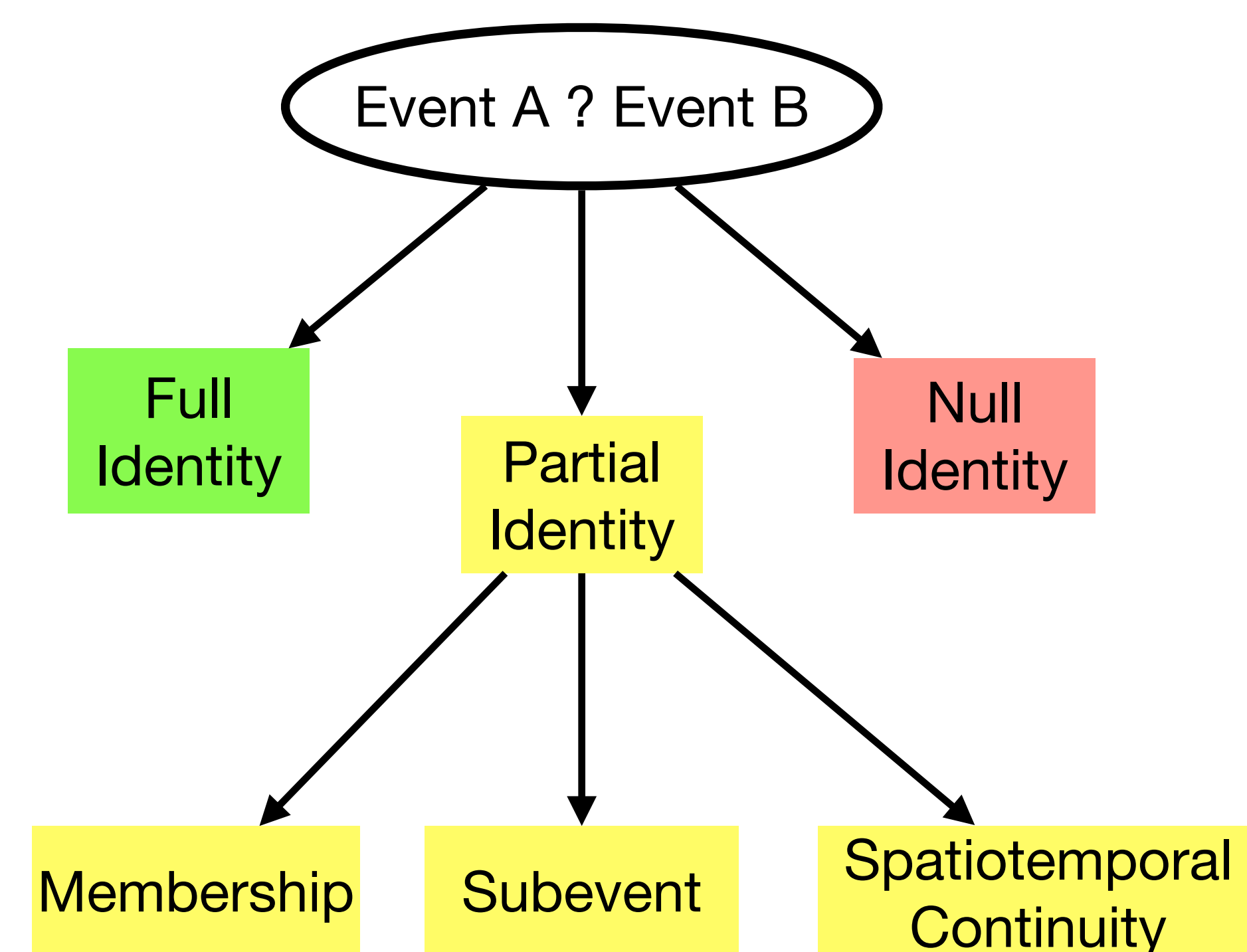


3. Baselines

- Two baseline models to predict coreference vs non-coreference
 - Lemma match between mentions
 - BERT-based cross-encoder

Baseline	Dev			Test		
	P	R	F1	P	R	F1
Lemma-match	46.6	54.9	49.9	42.3	56.0	48.2
Cross encoder	43.1 ±0.6	75.4 ±0.5	54.3 ±0.5	45.9 ±0.8	77.3 ±1.1	57.6 ±0.6

4. Partial Identity



Nearly 200 dead in Haitian cholera outbreak

Nearly 200 people are confirmed dead and approximately 2600 are ill in a central Haitian cholera **outbreak**, according to the Centers for Disease Control (CDC), ... Haitian officials place the death toll at 194 deaths with 2,364 people being infected.

'Explosive' Haitian cholera outbreak kills 292, neighboring countries prepare

The Haitian cholera **outbreak** has killed 292 people and infected over 4000, according to the Haitian government, ...

October 22, 2007

Membership

October 24, 2007

Reports say that the amount of people fleeing from their homes in California located in the United States due to **wildfires** has reached the 1,000,000 mark as the fires continue to grow.

October 26, 2010

Over 250 dead in Haiti cholera outbreak, thousands infected

At least 259 people are dead and over 3000 people have been infected in the Haitian cholera **outbreak**.

October 28, 2010

Spatiotemporal Continuity

July 18, 2007

The **fire** has burned about 4400 acres so far and 15 homes have been lost, however there have been no reported injuries or deaths.

Subevent

July 21, 2007

A freight train in Lviv, Ukraine derailed, caught fire, and spilled a toxic chemical, releasing dangerous fumes into the air early Tuesday morning (local time), and people who live near the site of the **crash** are still becoming sick.

The available information about the phosphorous cloud following the railway **accident** in the Ukraine last Monday is becoming more and more cryptic.